Light

GCSE Computer Science — Network Protocols (Set 2)

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Network Protocols

Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

An online shop uses a protocol that encrypts customer payment details during transmission.
Which protocol provides this secure connection?

- A. HTTPS
- B. FTP
- C. POP3
- D. SMTP

Q2 (1 mark)

A business user needs to access their emails from multiple devices while keeping all messages stored on the server.
Which protocol should they use?

- A. HTTP
- B. FTP
- C. IMAP
- D. Ethernet

Q3 (1 mark)

A graphic designer needs to transfer large image files from their computer to a client's server.
Which protocol is designed for this type of file transfer?

- A. SMTP
- B. HTTP
- C. FTP
- D. POP3

Q4 (1 mark)

A student types 'www.example.com' into their web browser.
Which protocol does the browser use to request the web page?

- A. HTTP
- B. SMTP
- C. POP3
- D. IMAP

Q5 (1 mark)

When a user clicks 'Send' on a new email, a protocol transmits the message to the outgoing mail server.
Which protocol handles this?

- ○ **A.** SMTP
- ○ **B.** HTTP
- ○ **C.** FTP
- ○ **D.** Ethernet

Q6 (1 mark)

An email protocol retrieves messages from a server and stores them on a single device.
Which protocol works this way?

- ○ **A.** HTTP
- ○ **B.** SMTP
- ○ **C.** POP3
- ○ **D.** FTP

Q7 (1 mark)

A smartphone connects to the internet through a wireless router at home.
Which technology enables this wireless connection?

- ○ **A.** Ethernet

- B. Wi-Fi
- C. SMTP
- D. FTP

Q8 (1 mark)

Desktop computers in an office are connected to the network using cables plugged into wall sockets.
Which technology is used for this wired connection?

- A. Wi-Fi
- B. IMAP
- C. HTTP
- D. Ethernet

Q9 (1 mark)

The TCP/IP model has a layer that handles protocols like HTTP, SMTP, and FTP.
What is this layer called?

- A. Link layer
- B. Internet layer
- C. Transport layer
- D. Application layer

Q10 (1 mark)

A TCP/IP layer uses port numbers to identify which application should receive incoming data.
Which layer is this?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Internet layer
- ○ **D.** Link layer

Q11 (2 marks)

A student learns that TCP/IP is essential for internet communication.
What is the role of TCP/IP?

- ○ **A.** It displays web pages in a browser window
- ○ **B.** It provides the fundamental rules for packaging, addressing, transmitting, and receiving data across networks
- ○ **C.** It stores files permanently on web servers
- ○ **D.** It creates the content of websites and emails

Q12 (2 marks)

Data needs to travel from one network to another across the internet.
Which TCP/IP layer handles addressing and routing for this journey?

- ○ **A.** Internet layer
- ○ **B.** Application layer

- ○ **C.** Transport layer
- ○ **D.** Link layer

Q13 (2 marks)

A switch forwards data frames within a local network using hardware addresses.
At which TCP/IP layer does this operation occur?

- ○ **A.** Link layer
- ○ **B.** Internet layer
- ○ **C.** Transport layer
- ○ **D.** Application layer

Q14 (2 marks)

A company's network administrator insists that all web traffic must use HTTPS instead of HTTP.
What security benefit does this provide?

- ○ **A.** Websites load faster because HTTPS uses compression
- ○ **B.** Users no longer need to remember passwords
- ○ **C.** All data transmitted between browsers and servers is encrypted, preventing eavesdropping
- ○ **D.** Viruses cannot infect computers using HTTPS

Q15 (2 marks)

An email travels from sender to recipient through multiple servers.
Which protocol is used for transmission between mail servers?

- ○ **A.** POP3
- ○ **B.** IMAP
- ○ **C.** HTTP
- ○ **D.** SMTP

Q16 (2 marks)

When sending data, each TCP/IP layer adds its own header information before passing data down.
What is this process called?

- ○ **A.** Decryption
- ○ **B.** Decompression
- ○ **C.** Authentication
- ○ **D.** Encapsulation

Q17 (2 marks)

A network uses both Ethernet cables and Wi-Fi connections.
At which TCP/IP layer do both Ethernet and Wi-Fi operate?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Internet layer

○ **D.** Link layer

Q18 (2 marks)

A user configures their new phone to access an existing email account that already works on their laptop.
If using POP3, what problem might occur?

○ **A.** The phone cannot send any emails

○ **B.** Emails downloaded to the laptop may not appear on the phone because they were removed from the server

○ **C.** The email account will be automatically deleted

○ **D.** The phone will receive duplicate copies of every email

Q19 (2 marks)

A teacher explains that protocols are essential for network communication.
Why do networks need standardised protocols?

○ **A.** To ensure devices from different manufacturers can communicate using agreed rules

○ **B.** To make networks run on less electricity

○ **C.** To prevent users from accessing certain websites

○ **D.** To store data more efficiently on servers

Q20 (2 marks)

The Transport layer offers two main protocols: TCP and UDP.

Which statement correctly describes when UDP is preferred over TCP?

- ○ **A.** UDP is preferred when downloading important files that must be complete
- ○ **B.** UDP is preferred for real-time applications like video calls where speed matters more than perfect delivery
- ○ **C.** UDP is preferred for online banking where every transaction must be verified
- ○ **D.** UDP is preferred for sending emails that must arrive without errors

Q21 (2 marks)

A security audit reveals that a company is using standard FTP for file transfers.
What security concern does this raise?

- ○ **A.** FTP transfers files too slowly for business use
- ○ **B.** FTP cannot transfer files larger than 10 MB
- ○ **C.** FTP transmits usernames, passwords, and file contents in plain text that can be intercepted
- ○ **D.** FTP only works on Windows computers

Q22 (2 marks)

A student is confused about addressing at different TCP/IP layers.
Which statement correctly matches address types to their layers?

- ○ **A.** MAC addresses are used at the Internet layer; IP addresses are used at the Link layer

- ○ **B.** Port numbers are used at the Application layer; IP addresses are used at the Transport layer
- ○ **C.** IP addresses are used at the Internet layer; MAC addresses are used at the Link layer
- ○ **D.** All layers use the same addressing system for simplicity

Q23 (2 marks)

A web developer creates a form for users to submit personal information.
Why should this form be on an HTTPS page rather than HTTP?

- ○ **A.** HTTPS forms can collect more data than HTTP forms
- ○ **B.** HTTPS encrypts the submitted data, protecting personal information during transmission
- ○ **C.** HTTP forms cannot be submitted to servers
- ○ **D.** HTTPS pages load faster than HTTP pages

Q24 (2 marks)

When receiving data, a computer processes information from the lowest TCP/IP layer upward.
What is the correct order of layers from bottom to top?

- ○ **A.** Application, Transport, Internet, Link
- ○ **B.** Link, Internet, Transport, Application
- ○ **C.** Transport, Link, Application, Internet

- ○ **D.** Internet, Link, Transport, Application

Q25 (2 marks)

A company uses multiple protocols for different network services. Which row correctly matches each protocol to its primary purpose?

- ○ **A.** HTTP: file transfer, SMTP: web browsing, FTP: email sending
- ○ **B.** HTTP: email receiving, SMTP: file transfer, FTP: web browsing
- ○ **C.** HTTP: web browsing, SMTP: email sending, FTP: file transfer
- ○ **D.** HTTP: email sending, SMTP: web browsing, FTP: email receiving

Q26 (3 marks)

A network administrator is troubleshooting a connectivity problem. They observe that:

• The computer can ping other devices by IP address successfully
• The computer cannot access any websites by domain name
• The email client cannot connect to mail.company.com

Based on this evidence, at which TCP/IP layer is the problem most likely occurring?

- ○ **A.** Link layer, because the network cable must be damaged
- ○ **B.** Internet layer, because IP routing is not working
- ○ **C.** Transport layer, because TCP connections are being blocked
- ○ **D.** Application layer, because DNS name resolution is failing

Q27 (3 marks)

A company is setting up email for employees and must decide between POP3 and IMAP.
The employees:
• Use multiple devices (phones, laptops, tablets)
• Need to access emails while travelling
• Want all devices to show the same email status
Which configuration should be chosen and why?

- ○ **A.** POP3, because it downloads emails faster than IMAP
- ○ **B.** POP3, because it automatically synchronises emails across all devices
- ○ **C.** IMAP, because it keeps emails on the server allowing consistent access from multiple devices with synchronised status
- ○ **D.** Either protocol works equally well for these requirements

Q28 (3 marks)

A web browser requests a page from a web server on a different network.
Describe the correct sequence of layer involvement when sending this request.

- ○ **A.** Link creates the request, Internet encrypts it, Transport adds web addresses, Application transmits it physically
- ○ **B.** Application formats the HTTP request, Transport adds port numbers and ensures reliability, Internet adds IP addresses for routing, Link creates frames for physical transmission
- ○ **C.** Transport initiates the web request, Application routes between networks, Link adds IP addresses, Internet handles MAC addresses
- ○ **D.** Internet creates the web page, Application adds physical addresses, Transport handles routing, Link manages port numbers

Q29 (3 marks)

A school network currently uses:
• HTTP for the student portal
• FTP for teachers uploading resources
• Open Wi-Fi with no password
A security consultant reviews these practices. What are the risks?

- ○ **A.** There are no significant risks as these are standard protocols
- ○ **B.** The only risk is slow performance due to outdated protocols
- ○ **C.** All three practices expose data to interception: HTTP transmits login credentials in plain text, FTP exposes teacher files and passwords, open Wi-Fi allows anyone to capture all network traffic
- ○ **D.** The risk is that too many students can connect at once

Q30 (3 marks)

A network engineer must choose between TCP and UDP for different applications:
• Application A: Online banking transactions
• Application B: Live sports video streaming
• Application C: File backup to cloud storage
Which protocol assignments are correct and why?

- ○ **A.** All three should use UDP because it is faster than TCP
- ○ **B.** All three should use TCP because reliability is always more important than speed
- ○ **C.** A and C should use TCP (reliability essential for transactions and complete file backup); B should use UDP (low latency more important than perfect delivery for live video)
- ○ **D.** A and B should use UDP (speed matters most); C should use TCP (files need reliability)

Total: **55** marks • Time: **30 min**

↑ Back to top

Submit answers

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.